

Digital Image Processing (CS-321)

Lecture:2

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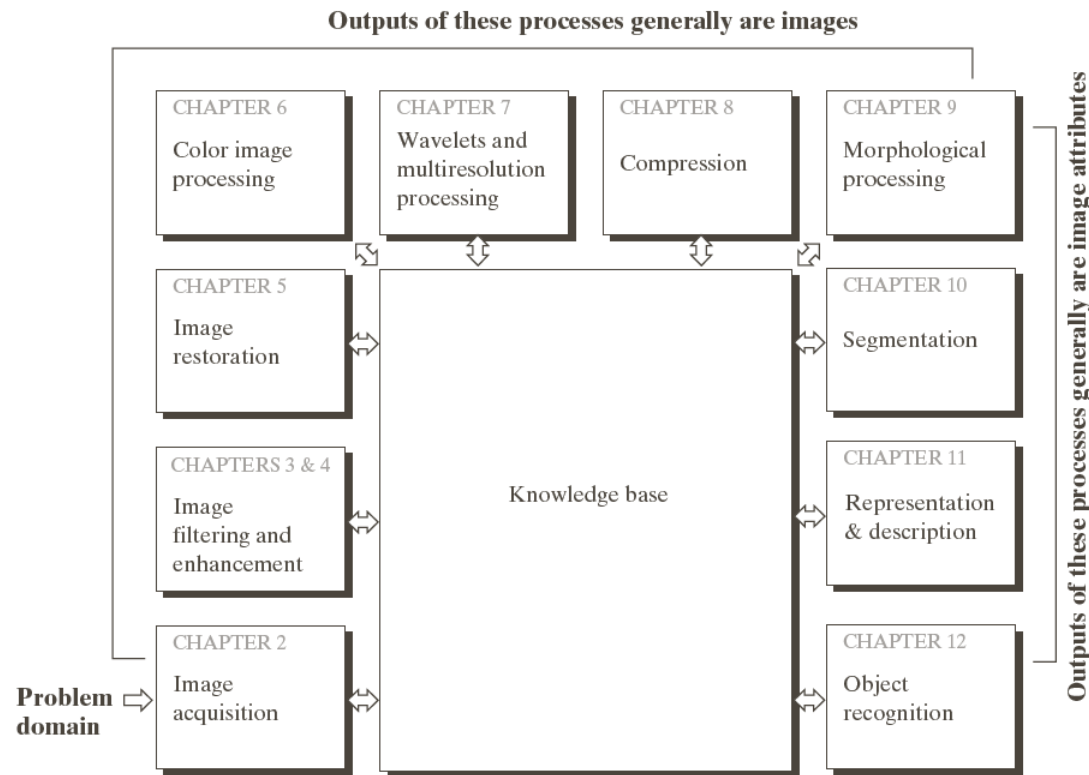
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- Topics to Learn

1. Fundamental steps in Image Processing
2. Elements of an Image Processing System

Fundamental Steps in Image Processing

- The steps are categorized on the basis of output form.
- As the input is image then output can be either **image** or **attributes of images**.



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Fundamental Steps in Image Processing

- For output as image- the steps are:
- **Image acquisition:**
 - It involves retrieving the image from a source usually a **hardware-based source**.
 - This step is also known as **preprocessing** in image processing. E.g. scaling
- **Image filtering & enhancement:**
 - It is the process of **manipulating an image** so that the result is more suitable than the original for a specific application.
 - The term '**specific**' is user application oriented and an enhancement technique that is more suitable for X-ray images may not be enough for satellite images.

- **Image restoration:**

- It is an area that also deals with improving the appearance of an image.
- However, unlike enhancement, which is subjective, image restoration is objective, in the sense that restoration techniques tends to based on mathematical or probabilistic models of image degradation.
- Enhancement on the other hand, is based on human subjective preference regarding what constitutes a “good” enhancement result.

- **Color Image Processing:**

- In color image processing, an abstract mathematical model known as **color space** is used to characterized the colors in term of intensity values.
- In RGB color, the red, green and blue use 8 bits each, which have integer values from 0 to 255. This makes $256*256*256=16777216$ possible colors.

- **Wavelets:**

- It provided the basic foundation to represent images in various **degrees of resolution**.
- Basically, a **pyramidal representation** is formulated in which the images are sub divided successively into smaller regions.

Fundamental Steps in Image Processing

- **Compression:**

- It deals with the techniques for **reducing the storage** required to save an image , or the bandwidth required to transmit it.
- For example- **JPEG** (Joint Photographic Expert Group).

- **Morphological Processing:**

- It deals with tool for extracting image components that are useful in the representation and description of **shape**.

- **Segmentation:**

- This procedure involves **partition of an image** into its subsequent parts or objects.

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Fundamental Steps in Image Processing

- **Representation and Description:**

- It almost always follow the output of a segmentation stage, which is usually a raw pixel data, constituting either the boundary of region or all the points in the region itself.
- These data can be a good representation and open the area for research in feature selection.

- **Recognition:**

- It is the process that assigns a label to an object based on its descriptors.

- **Knowledge base:**

- Knowledge about the domain problem is coded into an image processing system in the form of knowledge database.
- It can help to details the regions of the image.
- Also, it controls the interaction between the modules.

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All modules are in close proximity with the knowledge base.

Elements of Digital Image Processing System

- The basic components comprising a typical general-purpose system used for digital image processing is shown below.

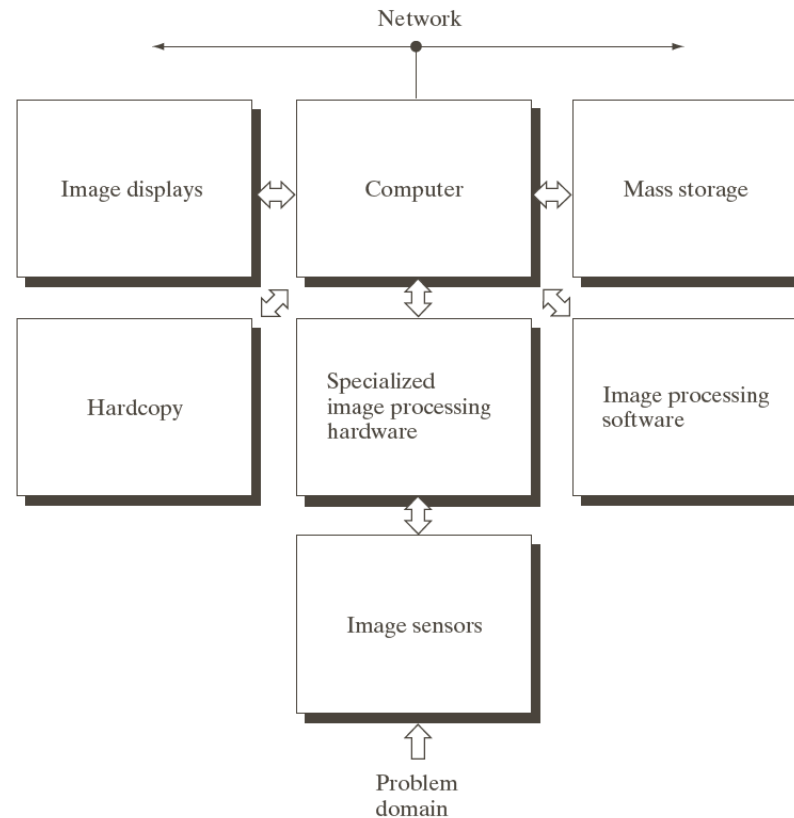


FIGURE 1.24
Components of a
general-purpose
image processing
system.

Elements of Digital Image Processing System

- For sensing, two elements are required:
 1. A **physical device** that is sensitive to energy radiation by the object we wish to image.
 2. **Digitizer**- A device that convert the output of the physical sensing device into digital form.
- **Specialized image processing Hardware**
 - It consists of **digitizer and hardware** that performs primitive operations such as **Arithmetic Logic Unit (ALU)**, that perform arithmetic and logical operations in parallel on entire images.
 - In other words, this unit performs functions that require **fast data throughput** (e.g.- digitizer and averaging videos frames at 30 frames/ sec) that typically main computer cannot handle.

Elements of Digital Image Processing System

- **Computer:** This is a general purpose PC or a supercomputer based on the level of performance required.
- **Software:** A well- designed package that includes the capabilities for the user to write code that, as a minimum, utilizes the specialized modules.
- **Mass Storage:** This capability is must for image processing application.
 - An image of size **1024X1024 pixels** in which the intensity of each pixel is an 8-bit quantity requires **one megabytes** storage.
 - When deal with large number of images, it is crucial to have enough storage.

Elements of Digital Image Processing System

- **Image display:** This provide/ show the color of images and monitors are driven by the output of the image and graphics display cards that are an integral part of the computer system.
- **Hardcopy:** It constitute recording images, using laser printer, film cameras, inkjet units, etc.
- **Networking:** It is the default function in any computer system due to large volume of inherent data of image processing applications and consideration is based on the bandwidth.